Document Investigation Dossier - pdf_slips_and_bank

Generated by DocForensics. Visual findings are indicators of possible alteration, not proof of fraud. Sections follow the investigation brief (7.1 to 7.6).

Case at a glance

Documents	Findings	High	Medium	Low
3	7	5	1	1

- 3 page(s) from 3 file(s); 3 recognised as a supported type, 0 not recognised.

- Text: 3 page(s) read exactly from the PDF text layer, 0 page(s) read by OCR.

- 23 field(s) extracted, 1 table(s) found.

- Visual forensics: pixel-level checks on 0 page(s), PDF structure checks on 3 generated PDF page(s).

- Cross-document: 8 field(s) compared across documents (7 agree, 1 differ, 0 in one document only, 0 could not be compared); 3 document pair(s) linked by shared identifiers.

- 7 finding(s) about the documents and 2 coverage note(s) (fields that could not be read, checks that did not apply).

7.1 Document Overview

Document	Type	Source	Processing status	SHA-256 (first 16)
bank_statement_conflict	bank_statement (class. conf. 0.6; Bank statement 0.81)	bank_statement_conflict.pdf (PDF, 2222 bytes), page 1 of 1, 1654x2339 px	recognised as bank statement; text read exactly from the PDF text layer; 5 field(s); PDF structure checks ran; pixel analysis not applicable	25686adff335ec33
salary_slip_clean	salary_slip (class. conf. 0.79; Salary slip 0.38)	salary_slip_clean.pdf (PDF, 2335 bytes), page 1 of 1, 1654x2339 px	recognised as salary slip; text read exactly from the PDF text layer; 9 field(s); PDF structure checks ran; pixel analysis not applicable	3fe744c8514db218
salary_slip_tampered	salary_slip (class. conf. 0.79; Salary slip 0.38)	salary_slip_tampered.pdf (PDF, 2518 bytes), page 1 of 1, 1654x2339 px	recognised as salary slip; text read exactly from the PDF text layer; 9 field(s); PDF structure checks ran; pixel analysis not applicable	20e25536d02e1a53

7.2 Extracted Information

Field values with the confidence they were read with (1.00 for text read exactly from a PDF text layer, OCR word confidence for scans and photos) and their location in the original page: [x0, y0, x1, y1] in pixels of the page image at 200 dpi (divide by 2.78 for PDF points). The document structure follows.

bank_statement_conflict (bank_statement)

Field	Value	Read conf.	Location (page px)
name	Rahul Verma Test	1.00	[344, 259, 568, 297]
account_no	XXXX XXXX 4821	1.00	[244, 304, 469, 342]
opening_balance	42,310	1.00	[1415, 465, 1500, 503]
period	01-Oct-2026 to 31-Oct-2026	1.00	[389, 181, 736, 219]
salary_credit	52,000	1.00	[1165, 554, 1250, 592]

salary_slip_clean (salary_slip)

Field	Value	Read conf.	Location (page px)
name	Rahul Verma Test	1.00	[556, 276, 779, 314]
employee_id	TST-10234	1.00	[556, 326, 694, 364]
account_no	XXXX 4821	1.00	[556, 476, 699, 514]
gross	70,000	1.00	[1026, 737, 1111, 775]
deductions	10,000	1.00	[1026, 781, 1111, 819]
net_pay	60,000	1.00	[1009, 831, 1111, 877]
id_number	ABCDE1234F	1.00	[556, 376, 730, 414]
pay_month	October 2026	1.00	[320, 181, 488, 219]
employer	ACME TEST SOLUTIONS PVT LTD (Fictional)	1.00	[139, 119, 1090, 180]

salary_slip_tampered (salary_slip)

Field	Value	Read conf.	Location (page px)
name	Rahul Verma Test	1.00	[556, 276, 779, 314]
employee_id	TST-10234	1.00	[556, 326, 694, 364]
account_no	XXXX 4821	1.00	[556, 476, 699, 514]
gross	70,000	1.00	[1026, 737, 1111, 775]
deductions	10,000	1.00	[1026, 781, 1111, 819]
net_pay	95,000	1.00	[1024, 834, 1115, 878]
id_number	ABCDE1234F	1.00	[556, 376, 730, 414]
pay_month	October 2026	1.00	[320, 181, 488, 219]
employer	ACME TEST SOLUTIONS PVT LTD (Fictional)	1.00	[139, 119, 1090, 180]

Document structure: sections and tables

Sections are found from headings; tables from their header row. Values are read from the OCR word boxes; checks only use rows whose amounts could be read.

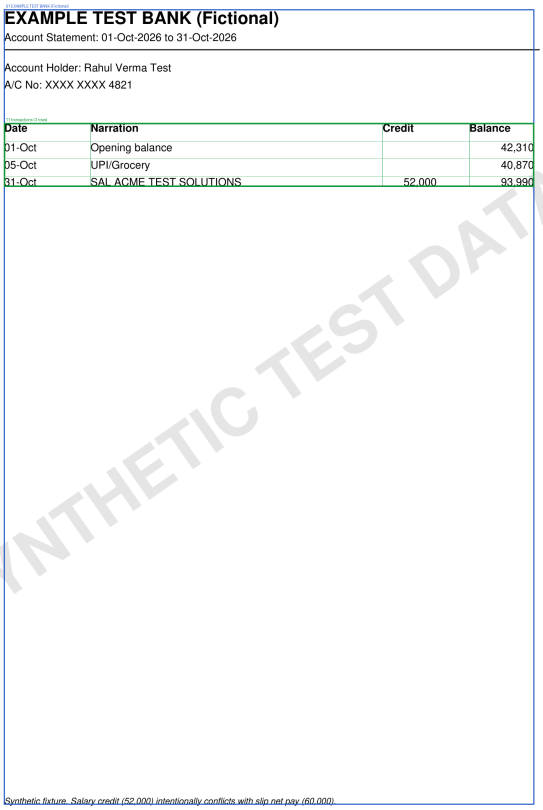
bank_statement_conflict (bank_statement)

Layout: statement: label-value header block + transaction table. Hierarchy: page > 1 section(s) > 4 field(s), 1 table(s) (3 rows, 10 cells). Evidence: 3 labelled field(s); 0 label-free field(s); 1 heading(s); transaction table with 3 rows.

Section	Title	Kind	Fields inside	Tables
S1	EXAMPLE TEST BANK (Fictional)	title	name, account_no, period	T1

Table T1: transactions, 3 rows; running balance checked on 1 rows, 1 break(s), 1 row(s) with unreadable amounts.

#	Date	Description	Withdrawal	Deposit	Balance
1	2026-10-01	Opening balance			42,310.00
2	2026-10-05	UPI/Grocery			40,870.00
3	2026-10-31	SAL ACME SOLUTIONS		52,000.00	93,990.00



EXAMPLE TEST BANK (Fictional)

Account Statement: 01-Oct-2026 to 31-Oct-2026

Account Holder: Rahul Verma Test

A/C No: XXXX XXXX 4821

Date	Narration	Credit	Balance
01-Oct	Opening balance		42,310
05-Oct	UPI/Grocery		40,870
31-Oct	SAL ACME TEST SOLUTIONS	52,000	93,990

SYNTHETIC TEST DATA

Synthetic fixture. Salary credit (52,000) intentionally conflicts with slip net pay (60,000).

salary_slip_clean (salary_slip)

Layout: label-value form. Hierarchy: page > 1 section(s) > 8 field(s), 0 table(s) (0 rows, 0 cells). Evidence: 6 labelled field(s); 1 label-free field(s); 1 heading(s).

Section	Title	Kind	Fields inside	Tables
S1	ACME TEST SOLUTIONS PVT LTD (Fictional)	title	name, employee_id, account_no, gross, deductions, net_pay, id_number, pay_month	-

salary_slip_tampered (salary_slip)

Layout: label-value form. Hierarchy: page > 1 section(s) > 8 field(s), 0 table(s) (0 rows, 0 cells). Evidence: 6 labelled field(s); 1 label-free field(s); 1 heading(s).

Section	Title	Kind	Fields inside	Tables
S1	ACME TEST SOLUTIONS PVT LTD (Fictional)	title	name, employee_id, account_no, gross, deductions, net_pay, id_number, pay_month	-

7.3 Contextual Analysis

How to read the status: potential alteration = an indicator that something may have been changed, not proof; inconsistency = a confirmed data conflict between values in the documents, which says nothing about intent; observation = a note, coverage gap or weak signal. Nothing here is a finding of fraud.

Identified field-level inconsistencies

Rules that must hold inside one document: arithmetic, totals and running balances, formats, checksums, date logic.

C-001 [HIGH | confidence high (0.98) | inconsistency] Salary slip arithmetic does not balance in salary_slip_tampered

Explanation: Gross 70,000 - deductions 10,000 = 60,000, but Net Pay shows 95,000 (difference +35,000). Corroborated by visual evidence (V-001) on the same field.

Evidence: salary_slip_tampered[20e25536].gross='70,000'; salary_slip_tampered[20e25536].deductions='10,000'; salary_slip_tampered[20e25536].net_pay='95,000' *Method:* arithmetic check

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Figures printed on the same page do not add up, so at least one of them is wrong or was changed after the document was produced.

Suggested check: Ask the issuer for the original or a recalculation and compare the figures with the payroll record.







T-001 [HIGH | confidence high (0.85) | inconsistency] Running balance does not reconcile in bank_statement_conflict (1 of 1 rows)

Explanation: Each row's balance must equal the previous balance plus deposits minus withdrawals. row 3 (2026-10-31): previous balance 40,870.00 + credit 52,000.00 = 92,870.00, but the statement prints 93,990.00 (difference +1,120.00) A single altered amount or balance breaks this chain.

Evidence: bank_statement_conflict[25686adf].T1 row 3 balance=93990.0 *Method:* table running-balance check

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: A value breaks a rule that should hold within this document (format, date logic, balance or total).

Suggested check: Verify the value against the original document and the issuing authority.



Missing or duplicated information

Expected fields that were absent or unreadable, the same field printed twice with different values, and values that sit under the wrong label.

T-003 [LOW | confidence medium (0.70) | observed] Pages salary_slip_clean and salary_slip_tampered are near-identical

Explanation: The printed content of the two pages overlaps 100% (different pages score below 80%, even with the same layout): the same document may have been submitted twice, or a page was copied. A copy with one value changed also scores this high.

Evidence: salary_slip_clean[3fe744c8] (whole page); salary_slip_tampered[20e25536] (whole page) *Method:* duplicate page check

Status: Observation: a note, coverage gap or weak signal, not an accusation

Why it needs a look: A value breaks a rule that should hold within this document (format, date logic, balance or total).

Suggested check: Verify the value against the original document and the issuing authority.

No region to highlight: this finding concerns the whole page or a value that could not be located.

Conflicting information across documents

Every field present in more than one document, with the value in each. 'agree' includes written-differently-but-compatible values (initials, omitted middle name, suffixes).

Field	Result	Values by document
Name	agree	bank_statement_conflict: Rahul Verma Test; salary_slip_clean: Rahul Verma Test; salary_slip_tampered: Rahul Verma Test
ID number	agree	salary_slip_clean: ABCDE1234F; salary_slip_tampered: ABCDE1234F
Employer	agree	salary_slip_clean: ACME TEST SOLUTIONS PVT LTD (Fictional); salary_slip_tampered: ACME TEST SOLUTIONS PVT LTD (Fictional)
Employee ID	agree	salary_slip_clean: TST-10234; salary_slip_tampered: TST-10234
Account number	agree	bank_statement_conflict: XXXX XXXX 4821; salary_slip_clean: XXXX 4821; salary_slip_tampered: XXXX 4821
Monthly gross income	agree	salary_slip_clean: 70,000; salary_slip_tampered: 70,000
Pay month	agree	salary_slip_clean: October 2026; salary_slip_tampered: October 2026
Payslip net pay vs bank payroll credit	DIFFER	salary_slip_clean: 60,000; salary_slip_tampered: 95,000; bank_statement_conflict: 52,000

Documents linked by shared identifiers: bank_statement_conflict - salary_slip_clean (account_no, name); bank_statement_conflict - salary_slip_tampered (account_no, name); salary_slip_clean - salary_slip_tampered (account_no, employee_id, id_number, name)

Cross-document discrepancies

X-001 [HIGH | confidence high (0.95) | inconsistency] Payslips for the same employee and month show different net pay

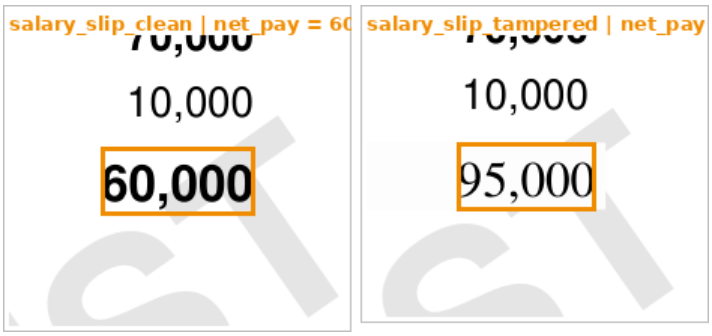
Explanation: salary_slip_clean: 60,000; salary_slip_tampered: 95,000 (employee TST-10234, 2026-10). Two payslips for one month should agree; either one was changed or they come from different sources. This is a data conflict to verify with the employer, not proof of fraud. Corroborated by visual evidence (V-001) on the same field.

Evidence: salary_slip_clean[3fe744c8].net_pay='60,000'; salary_slip_tampered[20e25536].net_pay='95,000' *Method:* same-month payslip comparison

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Documents that describe the same person or application disagree; at most one of them can be right.

Suggested check: Obtain the originals of the documents named, find which one matches the issuer's record, and do not assume which is correct.



X-003 [HIGH | confidence high (0.95) | inconsistency] Salary slip net pay does not match bank payroll credit

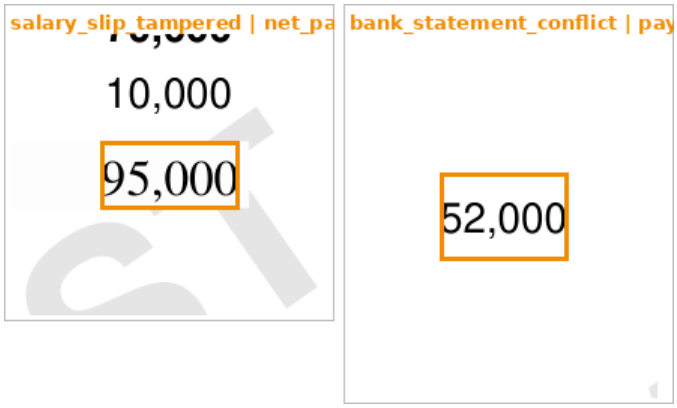
Explanation: Net pay on salary_slip_tampered is 95,000, but the payroll credit in the bank statement (bank_statement_conflict) on 2026-10-31 is 52,000 (difference +43,000). Either the payslip or the bank record is wrong; this is a data inconsistency to verify with the employer and the bank, not proof of fraud. Corroborated by visual evidence (V-001) on the same field.

Evidence: salary_slip_tampered[20e25536].net_pay='95,000'; bank_statement_conflict[25686adf].payroll_credit='52,000' *Method:* income reconciliation

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Records that should agree (payslip, bank statement, application, case reference) show different values.

Suggested check: Obtain the bank's own statement and the employer's payroll or confirmation record and see which side matches the issuer's data.



X-002 [HIGH | confidence high (0.90) | inconsistency] Salary slip net pay does not match bank payroll credit

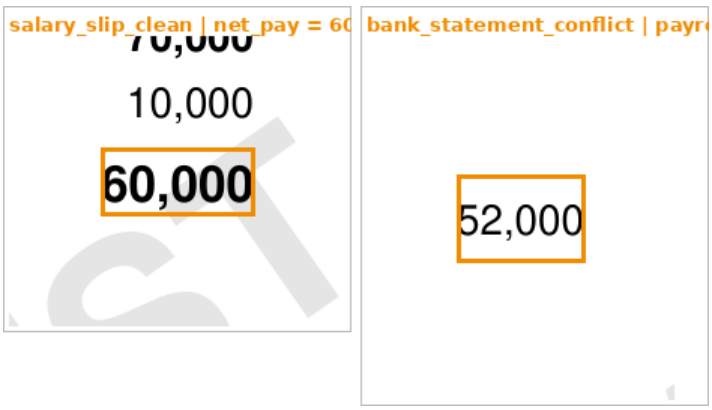
Explanation: Net pay on salary_slip_clean is 60,000, but the payroll credit in the bank statement (bank_statement_conflict) on 2026-10-31 is 52,000 (difference +8,000). Either the payslip or the bank record is wrong; this is a data inconsistency to verify with the employer and the bank, not proof of fraud.

Evidence: salary_slip_clean[3fe744c8].net_pay='60,000'; bank_statement_conflict[25686adf].payroll_credit='52,000' *Method:* income reconciliation

Status: Confirmed data conflict: two values in the submitted documents contradict each other (this says nothing about intent)

Why it needs a look: Records that should agree (payslip, bank statement, application, case reference) show different values.

Suggested check: Obtain the bank's own statement and the employer's payroll or confirmation record and see which side matches the issuer's data.



7.4 Visual Forensic Findings

Pixel-level and PDF-structure indicators that a region may differ from the rest of the page. They show where to look; they do not prove alteration. Each group says what was checked, so a group without findings states what was examined and found nothing.

Suspicious regions and potential alterations

V-001 [MEDIUM | confidence medium (0.69) | potential manipulation] Amount in a different font from the other amounts in salary_slip_tampered

Explanation: '95,000' is set in Times-Roman, while 5 other amounts on the page use helvetica. A figure typed over another in a different typeface can look like this. It can also be a deliberate emphasis; this is an indicator only, so compare with the original.

Evidence: salary_slip_tampered[20e25536].net_pay='95,000' *Method:* PDF structure: fonts

Status: Potential alteration: an indicator that something may have been changed, not proof

Why it needs a look: The way this PDF was built (fonts, drawing order, hidden layers, file history) differs from an untouched generated document.

Suggested check: Request the native or re-issued file from the issuer, inspect the region at full size and compare the two.



Font, alignment and structural inconsistencies

Checked: bank_statement_conflict: PDF structure checks (4 amounts compared (page font helvetica)) | salary_slip_clean: PDF structure checks (6 amounts compared (page font helvetica)) | salary_slip_tampered: PDF structure checks (6 amounts compared (page font helvetica))

None identified.

Image manipulation observations

Checked: bank_statement_conflict: not applicable (generated PDF, no scan or photo) | salary_slip_clean: not applicable (generated PDF, no scan or photo) | salary_slip_tampered: not applicable (generated PDF, no scan or photo)

None identified.

Signature-related observations

Checked: bank_statement_conflict: no signature label | salary_slip_clean: no signature label | salary_slip_tampered: no signature label

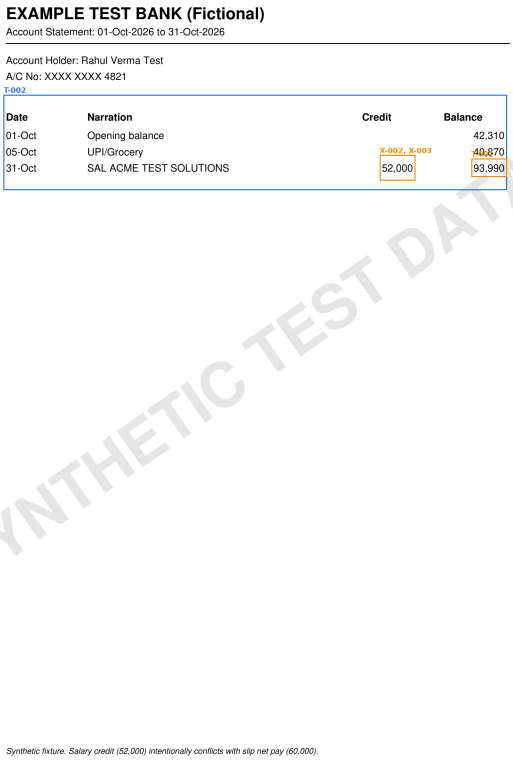
None identified.

7.5 Evidence and Traceability

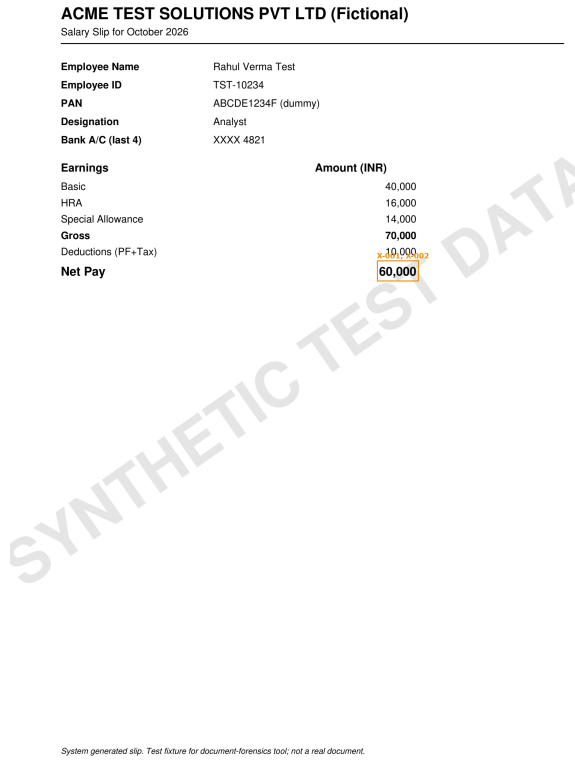
Each finding is tied to its original documents (file hash), the extracted fields and values it rests on, and the region it points to (box as in 7.2). Zoomed crops are printed with each finding in 7.3 and 7.4; the highlighted pages follow the table.

ID	Documents [SHA-256 first 8]	Associated extracted information	Region [x0,y0,x1,y1]	Status	Confidence
C-001	salary_slip_tampered[20e25536]	salary_slip_tampered.gross=70,000; salary_slip_tampered.deductions=10,000; salary_slip_tampered.net_pay=95,000	[1026, 737, 1111, 775]; [1026, 781, 1111, 819]; [1024, 834, 1115, 878]	inconsistency	high (0.98)
X-001	salary_slip_clean[3fe744c8], salary_slip_tampered[20e25536]	salary_slip_clean.net_pay=60,000; salary_slip_tampered.net_pay=95,000	[1009, 831, 1111, 877]; [1024, 834, 1115, 878]	inconsistency	high (0.95)
X-003	bank_statement_conflict[25686adf], salary_slip_tampered[20e25536]	salary_slip_tampered.net_pay=95,000; bank_statement_conflict.payroll_credit=52,000	[1024, 834, 1115, 878]; [1165, 544, 1250, 602]	inconsistency	high (0.95)
X-002	bank_statement_conflict[25686adf], salary_slip_clean[3fe744c8]	salary_slip_clean.net_pay=60,000; bank_statement_conflict.payroll_credit=52,000	[1009, 831, 1111, 877]; [1165, 544, 1250, 602]	inconsistency	high (0.90)
T-001	bank_statement_conflict[25686adf]	bank_statement_conflict.T1 row 3 balance=93990.0	[1415, 554, 1500, 592]	inconsistency	high (0.85)
V-001	salary_slip_tampered[20e25536]	salary_slip_tampered.net_pay=95,000	[1024, 842, 1115, 876]	potential manipulation	medium (0.69)
T-003	salary_slip_clean[3fe744c8], salary_slip_tampered[20e25536]	(whole page)	-	observed	medium (0.70)

Highlighted regions (boxes are labelled with finding IDs; colours: red potential alteration, orange inconsistency, blue observation)



bank_statement_conflict



salary_slip_clean

ACME TEST SOLUTIONS PVT LTD (Fictional)

Salary Slip for October 2026

Employee Name	Rahul Verma Test
Employee ID	TST-10234
PAN	ABCDE1234F (dummy)
Designation	Analyst
Bank A/C (last 4)	XXXX 4821

Earnings	**Amount (INR)**
Basic	40,000
HRA	16,000
Special Allowance	14,000
Gross	**70,000**
Deductions (PF+Tax)	10,000
Net Pay	95,000

SYNTHETIC TEST DATA

System generated slip. Test fixture for document-forensics tool; not a real document.

salary_slip_tampered

7.6 Investigation Summary

Case pdf_slips_and_bank: 3 documents analysed; 7 finding(s) about the documents (5 high, 1 medium, 1 low) and 2 coverage note(s). Most significant: C-001 Salary slip arithmetic does not balance in salary_slip_tampered; X-001 Payslips for the same employee and month show different net pay; X-003 Salary slip net pay does not match bank payroll credit; X-002 Salary slip net pay does not match bank payroll credit. Visual findings are indicators of possible alteration, not proof; confirm against originals. Pixel-level visual forensics ran on 0 of 3 recognised page(s); PDF structure checks (fonts, overlays, hidden text, file history) ran on 3 generated PDF page(s), where pixel analysis does not apply.

Summary source: template

Identified anomalies

ID	Anomaly	Severity	Status	Confidence	Documents
C-001	Salary slip arithmetic does not balance in salary_slip_tampered	high	inconsistency	high (0.98)	salary_slip_tampered
X-001	Payslips for the same employee and month show different net pay	high	inconsistency	high (0.95)	salary_slip_clean, salary_slip_tampered
X-003	Salary slip net pay does not match bank payroll credit	high	inconsistency	high (0.95)	bank_statement_conflict, salary_slip_tampered
X-002	Salary slip net pay does not match bank payroll credit	high	inconsistency	high (0.90)	bank_statement_conflict, salary_slip_clean
T-001	Running balance does not reconcile in bank_statement_conflict (1 of 1 rows)	high	inconsistency	high (0.85)	bank_statement_conflict
V-001	Amount in a different font from the other amounts in salary_slip_tampered	medium	potential manipulation	medium (0.69)	salary_slip_tampered
T-003	Pages salary_slip_clean and salary_slip_tampered are near-identical	low	observed	medium (0.70)	salary_slip_clean, salary_slip_tampered

Supporting evidence

Every anomaly above rests on the references, zoomed crops and highlighted regions in 7.3 to 7.5, and on the hash-chained audit log (appendix C), which records the references, the SHA-256 of the source file and of the evidence image, and the processing events that produced them.

Potential explanations

Possible, non-exclusive explanations to rule in or out when checking the originals. They are hypotheses, not conclusions.

ID	Possible explanations
C-001	A typing or calculation error when the document was produced · One figure edited after the document was produced · A different version of the statement mixed with this one
X-001	A typing error in one of the documents · The documents were produced at different times or are different versions · A value in one document was changed · A document of another person was mixed in
X-003	Timing: the payslip was corrected or reissued after the bank credit · Net pay shown before an adjustment that the bank reflects · A figure on the payslip or in the bank record was changed
X-002	Timing: the payslip was corrected or reissued after the bank credit · Net pay shown before an adjustment that the bank reflects · A figure on the payslip or in the bank record was changed
T-001	A typing or calculation error when the document was produced · One figure edited after the document was produced · A different version of the statement mixed with this one
V-001	A legitimate correction re-issued by the issuer · A design element (badge, highlight) that looks like a patch · A value overwritten after the document was generated
T-003	A second party or column (for example year-to-date) printed next to the first · The page or row was copied or overwritten

Areas requiring additional verification

Area	Why it needs verification
3 generated PDF page(s): pixel analysis does not apply	Only file-structure checks were possible. A document altered before it was turned into a PDF, or issued with wrong values, passes them. Compare with the issuer's own copy.

Area	Why it needs verification
Issuer verification	No document was checked against the issuing organisation or a registry. Confirm identity, employment and account details with the issuers.
Original documents	Findings come from the submitted copies only. Compare every flagged value and region with the originals before any decision.

Investigation limitations

Peer-relative statistics need at least four extracted fields per page. Heavily re-compressed or re-scanned documents reduce visual-forensic sensitivity. OCR errors can create false content mismatches. Skilled forgeries that match font, ink, tone and keep all numbers consistent are not detected by pixel forensics. Generated PDFs get structure checks only. No document was verified with its issuer. All flagged items should be verified against the original documents and the issuing authority.

Confidence levels: high (0.80 and above), medium (0.60 to 0.79), low (below 0.60). The number is the system's own estimate of how reliable the specific check is on this evidence (for example an exact arithmetic mismatch scores high, a weak pixel signal scores low). It is not a probability that a document is forged, and it does not say which of two conflicting documents is correct.

Appendix A. Capabilities and limitations of this system

Capability	Status
Ingestion: images, PDFs and ZIP bundles (one zip = one case); multi-document cases	Implemented
Page normalisation: deskew (tilt up to +/-8 deg) and lighting flattening for phone photos	Implemented; perspective correction planned
OCR + field extraction with field-level confidence and bounding boxes	Implemented (Tesseract + optional RapidOCR; printed text, label:value layouts)
Semantic label matching (embeddings) for field labels outside the dictionary	Implemented (needs embedding model; falls back to fuzzy dictionary)
Document classification: embeddings + keywords, unsupported-class veto, structural gate (>=3 expected fields)	Implemented (keyword + structure fallback when no model)
Visual forensics: ELA, grain/noise, paper tone, ink darkness (peer-relative, lighting-corrected)	Implemented
Contextual checks: arithmetic, formats, date logic, completeness	Implemented
Cross-document entity matching (odd-one-out) and income reconciliation	Implemented
Section detection; transaction and earnings/deductions tables with running-balance, page-continuity, total and component-sum checks	Implemented (bank-statement and salary-slip layouts)
Evidence mapping (annotated regions), audit trail, reviewer UI, PDF/JSON dossier	Implemented
Handwriting recognition	Unsupported
Typography / spacing / alignment checks, JPEG block-grid consistency (inserted or replaced content), photo-region check on ID cards, signature detection with placement and background checks	Implemented (heuristic; see limitations)
Born-digital PDFs: exact text layer (no OCR), font consistency of amounts, bordered patches over values, covered or hidden text, file history (revisions, dates, producer, Info vs XMP records, font subset tags)	Implemented (observations about file structure, not proof)
Duplicated (copy-move) image regions on scans and photos; image metadata (editing software, modification after capture); scanned-PDF file history	Implemented (observations; copies recompressed below about JPEG quality 92 after pasting can be missed)
Signature authenticity, photo identity matching	Planned
Layout-aware ML models (LayoutLM etc.), tables in other layouts, multilingual OCR	Planned

Appendix B. Standards and reference frameworks (no compliance claim)

This system draws on the frameworks below as reference points. It does not claim compliance or certification with any of them; each row says what we do and what we do not do.

Framework	What we do	What we do not do
ISO/IEC 27037 (identification, collection, acquisition and preservation of digital evidence)	Each source file is hashed (SHA-256) at ingest; originals are never modified (analysis uses derived copies, recorded in the log); every step is written to a hash-chained audit log with archived run history.	No acquisition or custody control before upload: the system only sees the copy it is given. No write protection or collection records for the original device or system.
ISO/IEC 27041 (suitability and adequacy of investigative methods)	Methods are documented (ARCHITECTURE.md), tested on a synthetic benchmark and a separately generated synthetic test pack with an answer key, and limitations are stated openly (skilled forgeries, tuning on synthetic data).	No formal validation study, no measured error rates on real documents, no independent testing.
ISO/IEC 27042 (analysis and interpretation of digital evidence)	Findings separate a data inconsistency, a potential alteration and an observation; each has a confidence level, possible explanations and a suggested check; nothing is called fraud; the pipeline is deterministic and repeatable.	Confidence levels are the system's own estimates, not calibrated probabilities; no formal competence or review process around the analyst.

Framework	What we do	What we do not do
NIST SP 800-86 (forensic techniques in incident response)	The pipeline follows its collection, examination, analysis and reporting phases: ingest and hashing; reading, extraction, structure and forensic checks; contextual and cross-document analysis; the dossier.	It is written for incident response on live systems; volatile data, network and system sources are out of scope here.
OWASP Machine Learning Security Top 10	No training is done and no model is hosted, so model theft, poisoning, inversion and membership inference do not apply. Supply chain: the downloaded embedding model is checked against recorded SHA-256 values and the result is logged. Output integrity: hash-chained audit and review logs. The optional Claude summary is off by default.	Input manipulation (documents crafted to mislead OCR or classification) is only partly mitigated (veto rule, minimum expected fields) and not tested adversarially. Recorded model hashes were taken from our own first download, not from a publisher signature. RapidOCR models are bundled in their package and not separately pinned. When the optional summary is on, finding text (document values) is sent to an external API.

Appendix C. Audit trail and investigation history

Hash-chained event log of run 20261105T170234-77d95c: 48 events, chain **verified intact** (checked in memory and as written to disk). Digest of the final event (SHA-256): 9bf70b4ad1f9bc68c0e249a08d04e3743a70ab00242d782bad3c1883d36b8c4f

Each finding record in the log carries its references, the SHA-256 of the source file and of the evidence crop, and the hashes of the ingest, OCR/text-layer, extraction and field-report events that produced it. References are shown as document[first 8 characters of the file's SHA-256]. Reviewer decisions are kept in review_log.jsonl (separate hash chain, anchored to this run's digest).

Earlier runs of this case: none (first run)

Time	Event	Detail
17:02:34	case_start	{'run': '20261105T170234-77d95c', 'case_id': 'pdf_slips_and_bank', 'n_inputs': 3, 'run_id': '20261105T170234-7
17:02:36	embedding_model	{'run': '20261105T170234-77d95c', 'backend': 'fastembed:BAAI/bge-small-en-v1.5', 'active': True, 'integrity':
17:02:36	ingest	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'source': 'bank_statement_conflict.pdf'
17:02:37	ingest	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'source': 'salary_slip_clean.pdf', 'sha256':
17:02:37	ingest	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'source': 'salary_slip_tampered.pdf', 'sha
17:02:37	ocr	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'engine': 'pdf-text-layer', 'n_words':
17:02:37	cell_fields	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'added': [('opening_balance', 'Opening
17:02:37	context_fields	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'added': ['period']}
17:02:37	extract	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'fields': ['name', 'account_no', 'openi
17:02:40	classify	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'doc_type': 'bank_statement', 'best_gue
17:02:40	ocr	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'engine': 'pdf-text-layer', 'n_words': 60, 'm
17:02:40	pattern_fields	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'added': ['id_number']}
17:02:40	context_fields	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'added': ['pay_month']}
17:02:40	extract	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'fields': ['name', 'employee_id', 'account_no
17:02:40	classify	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'doc_type': 'salary_slip', 'best_guess': 'sal
17:02:40	ocr	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'engine': 'pdf-text-layer', 'n_words': 60,
17:02:40	pattern_fields	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'added': ['id_number']}
17:02:40	context_fields	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'added': ['pay_month']}

Time	Event	Detail
17:02:40	extract	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'fields': ['name', 'employee_id', 'account
17:02:40	classify	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'doc_type': 'salary_slip', 'best_guess': '
17:02:40	categorise	{'run': '20261105T170234-77d95c', 'file': 'bank_statement_conflict.pdf', 'label': 'Bank statement', 'confidenc
17:02:40	categorise	{'run': '20261105T170234-77d95c', 'file': 'salary_slip_clean.pdf', 'label': 'Salary slip', 'confidence': 0.376
17:02:40	categorise	{'run': '20261105T170234-77d95c', 'file': 'salary_slip_tampered.pdf', 'label': 'Salary slip', 'confidence': 0.
17:02:40	structure	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'sections': [('EXAMPLE TEST BANK (Ficti
17:02:40	payroll_credits	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'credits': [('2026-10-31', 52000.0, 'AC
17:02:40	structure	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'sections': [('ACME TEST SOLUTIONS PVT LTD (F
17:02:40	structure	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'sections': [('ACME TEST SOLUTIONS PVT LTD
17:02:40	forensics_skipped	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'reason': 'born-digital PDF page (no sc
17:02:40	forensics_skipped	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'reason': 'born-digital PDF page (no scan/cap
17:02:40	forensics_skipped	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'reason': 'born-digital PDF page (no scan/
17:02:40	pdf_forensics	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'checks': {'amount_fonts': '4 amounts c
17:02:40	pdf_forensics	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'checks': {'amount_fonts': '6 amounts compare
17:02:40	pdf_forensics	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'checks': {'amount_fonts': '6 amounts comp
17:02:41	issuer_as_employer	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'employer': 'ACME TEST SOLUTIONS PVT LTD (Fic
17:02:41	issuer_as_employer	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'employer': 'ACME TEST SOLUTIONS PVT LTD (
17:02:41	fields_final	{'run': '20261105T170234-77d95c', 'doc_id': 'bank_statement_conflict', 'doc_type': 'bank_statement', 'source_s
17:02:41	fields_final	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_clean', 'doc_type': 'salary_slip', 'source_sha256': '
17:02:41	fields_final	{'run': '20261105T170234-77d95c', 'doc_id': 'salary_slip_tampered', 'doc_type': 'salary_slip', 'source_sha256'
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'V-001', 'kind': 'visual', 'severity': 'medium', 'confidence': 0.69,
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'C-001', 'kind': 'intra', 'severity': 'high', 'confidence': 0.98, 'st
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'X-001', 'kind': 'cross', 'severity': 'high', 'confidence': 0.95, 'st
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'X-002', 'kind': 'cross', 'severity': 'high', 'confidence': 0.9, 'sta
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'X-003', 'kind': 'cross', 'severity': 'high', 'confidence': 0.95, 'st
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'T-001', 'kind': 'intra', 'severity': 'high', 'confidence': 0.85, 'st
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'T-002', 'kind': 'missing', 'severity': 'low', 'confidence': 0.8, 'st
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'T-003', 'kind': 'intra', 'severity': 'low', 'confidence': 0.7, 'stat

Time	Event	Detail
17:02:42	finding	{'run': '20261105T170234-77d95c', 'fid': 'N-001', 'kind': 'missing', 'severity': 'low', 'confidence': 0.9, 'st
17:02:43	dossier	{'run': '20261105T170234-77d95c', 'case_id': 'pdf_slips_and_bank', 'n_findings': 9, 'summary_source': 'templat